

Architecture Specialist (ODC)

Practice Exam — 30 Questions | Time: 90 minutes

1. What is the main benefit of using Libraries in ODC?

- A.** They help implement business-agnostic components.
- B.** They can be strongly coupled with consumer apps for fast code execution.
- C.** They allow sharing code with apps or other libraries.
- D.** They allow packaging code to be included in the app's container.

2. Which of the following criteria is NOT key to choosing between a simple or distributed approach when assembling bounded concepts into apps?

- A.** Business Owner
- B.** Business Sponsor
- C.** Users expectations
- D.** Number of Product teams

3. What is the main objective of using a black box approach and encapsulating business data in a producer app?

- A.** To reduce the development effort for functionalities that require data mashup.
- B.** To ensure business rules associated with the data model are centralized.
- C.** To easily expose the business data model and rules to consumers.
- D.** To mitigate the impact of the breaking changes, led by the evolution of business rules, in the producer app.

4. How does OutSystems Developer Cloud's queuing mechanism contribute to the resilience of Event processing?

- A.** It enables Events to be recovered and replayed indefinitely until they run successfully, following their order in the queue.
- B.** It guarantees all Events are processed within the maximum execution time limit.
- C.** It ensures that Events are processed in the order they were triggered, providing predictable execution.
- D.** It temporarily holds Events' information until the subscriber app has available instances to process them.

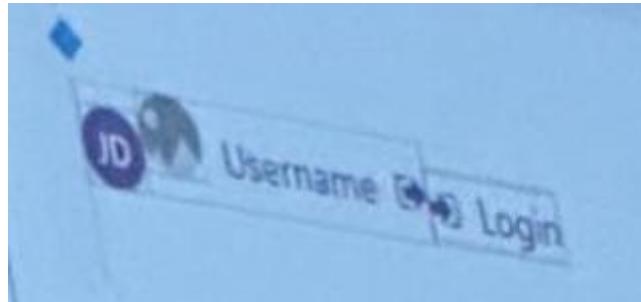
5. Which of the following scenarios describes a situation where there are strong references between different applications or Libraries?

- A.** Whenever an application references an element from another application.
- B.** All the elements from Libraries are strongly referenced by applications, except when referencing Service Actions.
- C.** Whenever an application references an element from a Library.
- D.** Whenever an application references a Server/Client Action from another application.

6. What is guaranteed about the order of Event delivery in OutSystems Developer Cloud?

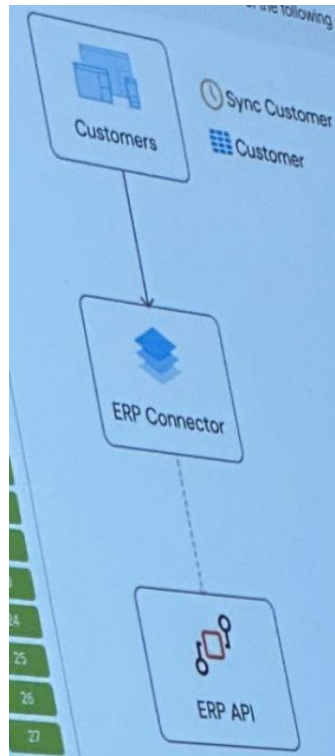
- A.** Events are delivered based on priority settings defined by the developer.
- B.** Events are delivered in the order they are triggered.
- C.** The delivery order of Events is not guaranteed.
- D.** Events are delivered strictly in a first-in, first-out (FIFO) order within the same event queue.

7. The image shows the user information in the Directory application. This UI is implemented using a Block called *EmployeeInformation*. A new app, called Vacations, must also share the same look and feel to display the same employee information. Knowing the necessary employee information is in the Directory app, what should you do to implement this scenario following the best practices to promote reusability?



- A.** Set the *EmployeeInformation* Block as Public. Then, reference the Block in the Vacations app directly, and add it to the Screens.
- B.** Create a new Block in the Vacations app with an Input parameter and similar UI to the *EmployeeInformation*, leveraging the Input Parameter as the data source. Create a Service Action in the Directory app that fetches and returns the data represented in the *EmployeeInformation* Block. In the Vacations app, reference the Service Action and pass the returned data to the Block via the input parameter.
- C.** Move the Block to a new Library and reference it from the two apps. Add an Input parameter to the Block and leverage it as the data source. Create a Service Action in the Directory app that fetches and returns the data represented in the *EmployeeInformation* Block. In the Vacation App, reference the Service Action and pass the returned data to the Block via the input parameter.
- D.** Create a new Block in the Vacations app with an Input parameter and similar UI to the *EmployeeInformation*, leveraging the Input Parameter as the data source. Reference the Entities in the Directory app that hold the data represented in the *EmployeeInformation* Block. Fetch the data directly from the Entities and pass it to the Block via the input parameter.

8. The scenario shows one application, Customers, connecting to an external ERP. The Customers app has an Entity that holds a local copy of the customer information and a Timer to synchronize the data from the third-party API to the local copy. Taking into account the scenario, which of the following options is correct?

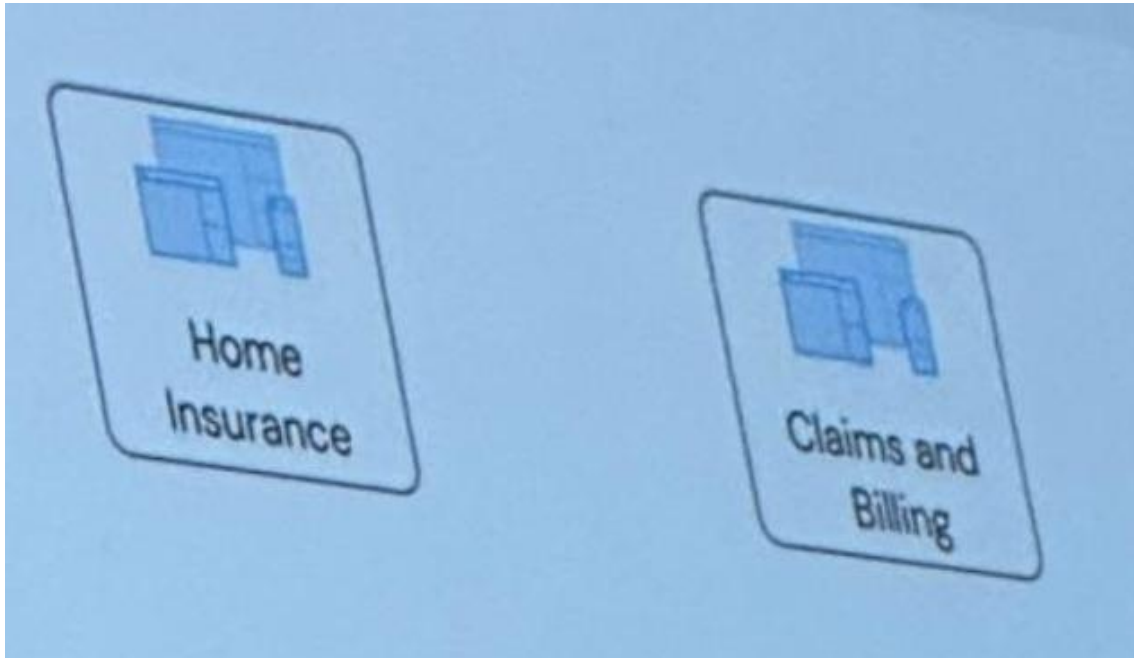


- A.** This approach is recommended when integrating with third-party APIs with high response times.
- B.** This approach triggers a data sync to the Customer Entity whenever a Customer is updated in the ERP system.
- C.** This approach supports real-time data integration with the third-party API.
- D.** This approach requires the third-party API to be able to execute a callback to a REST endpoint on the application.

9. Which of the following aspects should be considered during the 'Disclose' part of the Architecture Design Process?

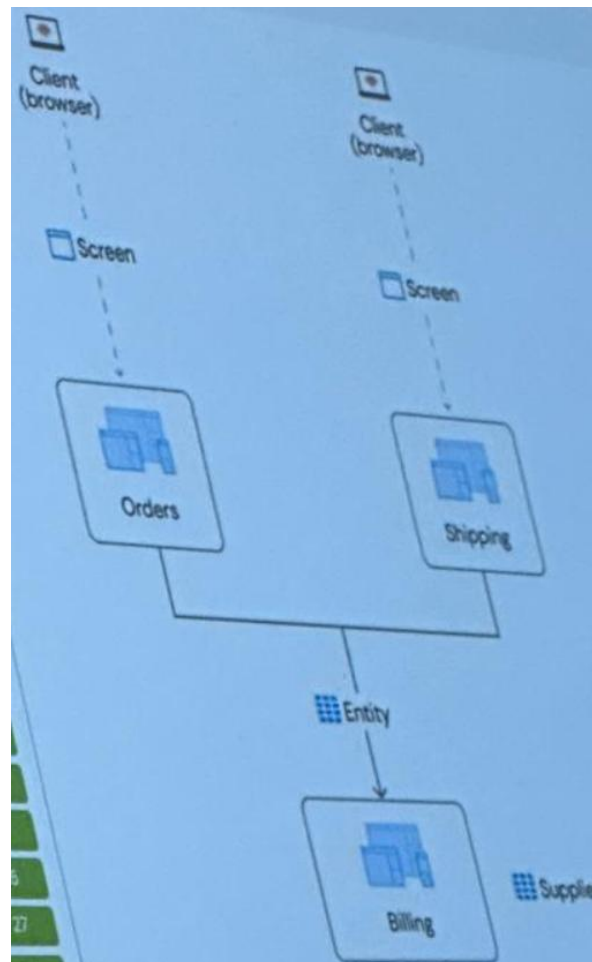
- A.** Release cycle independence
- B.** Personas and roles
- C.** Data flow and ownership
- D.** Business domain boundaries

10. Consider two applications that will share the same Theme and also important UI patterns to implement the Screens. What is the best approach to implement the Themes and Patterns, considering they can both be used in other applications?



- A.** Create one Library to implement the Theme and the Patterns. Then, both consumer applications can reference the elements directly from the Library.
- B.** Create one Library to implement the Theme and one application to implement the Patterns. Then, both consumer applications can reference the elements directly from the Library and Pattern application.
- C.** Create one Library to implement the Theme and another one to implement the Patterns. Then, both consumer applications can reference the elements directly from the Libraries.
- D.** Create one Library to implement the Theme and one Library to implement the Patterns. The Patterns Library references the Theme directly, and then both consumer applications can directly reference the Blocks that implement the Patterns.

11. Consider the following scenario where the Supplier Entity is being shared with the Orders and Shipping apps. The business data in the Supplier Entity consists of thousands of records, has a low change pace, and tends to remain stable as the business evolves. The consumer apps use Aggregates to fetch the data and perform calculations with it. According to the OutSystems best practices, what improvement would you make, if any, to the solution presented in the image?



- A.** Encapsulate the business data and respective functionality in Service Actions and make the Entity private. The consumer apps can then reference and use these Service Actions.
- B.** Encapsulate the business data and respective functionality in Server Actions and make the Entity private. The consumer apps can then reference and use these Server Actions.
- C.** Split the Supplier Entity information into multiple Entities, by type of Supplier, to avoid complex calculations over a long number of records.
- D.** No further improvements are needed. The solution already follows the recommended best practice.

12. Which of the following options describes a recommended best practice regarding naming conventions in the OutSystems Developer Cloud?

- A.** The naming convention followed by Libraries should consider the Apps that are using them to promote code maintainability.
- B.** Applying the naming conventions is mainly important for business owners, sponsors, and stakeholders to understand the purpose of the Apps and Libraries.
- C.** Naming conventions only need to be applied to apps, since library names do not impact the URL.
- D.** App's names should follow the name of the bounded context they represent.

13. Which of the following elements can NOT be included in a Library in ODC?

- A.** Service Actions
- B.** Integration Wrappers
- C.** Static Entities
- D.** Blocks and Patterns

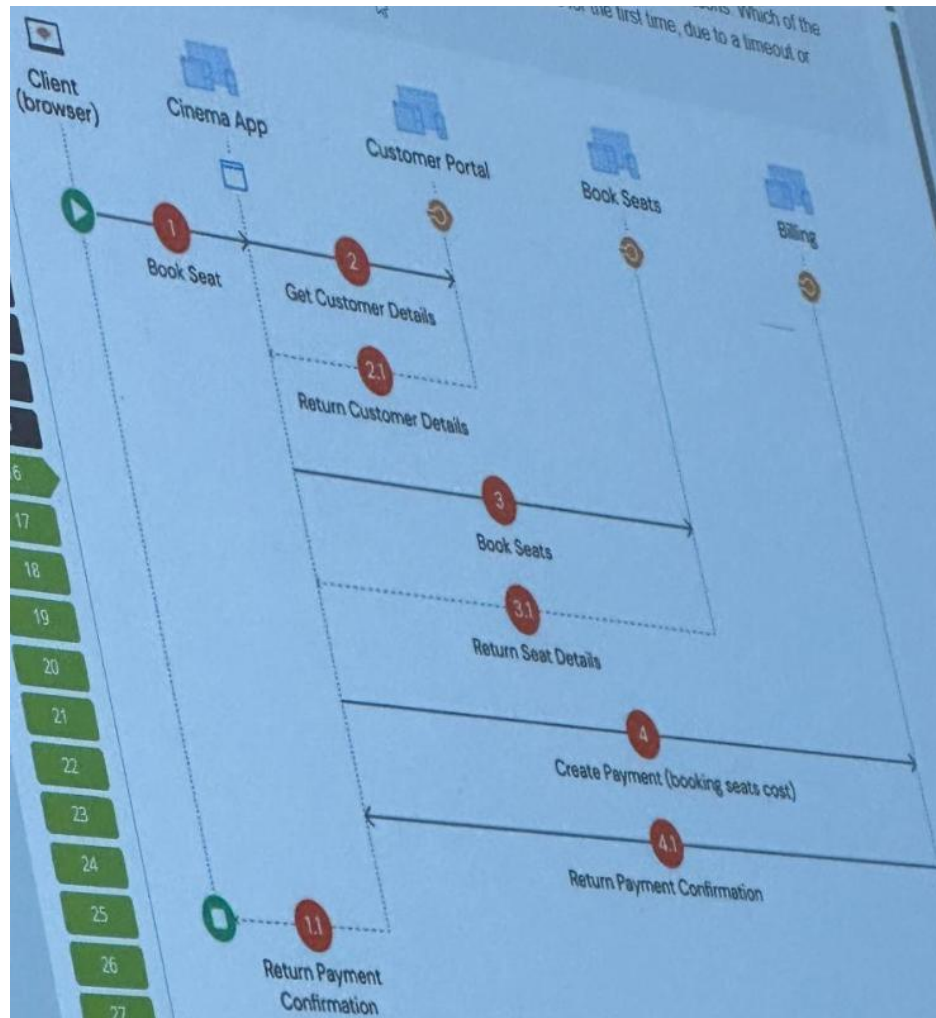
14. Which of the following options does NOT qualify as a compensation mechanism in case of a transaction failure?

- A.** Creating a reverse operation, like a debit to compensate for a wrongful credit.
- B.** Reverting back the application state to a consistent one.
- C.** Retrying the failed operation to recover the transaction.
- D.** Rolling back or soft-deleting data that was created.

15. Considering the Architecture Design Process, which of the following options is a goal of the 'Disclose' step?

- A.** Mapping business concepts to their respective teams
- B.** Identifying dependencies between business concepts
- C.** Understanding the business requirements and concepts
- D.** Grouping business concepts into bounded contexts

16. In the following scenario, the Payment operation from the Billing app can fail due to several reasons. Which of the following approaches should be followed when the payment operation fails for the first time, due to a timeout or connection issue?



- A. Retry the payment operation to try to successfully recover the transaction.
- B. Leave the payment pending and create a Timer that periodically retries the operation until the transaction is finally complete.
- C. Follow the built-in OutSystems mechanism to handle transaction failures.
- D. Create logic to roll back all the other operations and delete any records that those operations might have created in the database.

17. Having clear ownership can bring several benefits when assembling an Architecture Design. Which of the following options describes a reason for that?

- A.** It decreases complexity
- B.** It reduces cognitive load
- C.** It increases the communication flow
- D.** It helps to understand the organization

18. Which of the following sentences regarding Timers in ODC is correct?

- A.** A Timer can only be created in an application.
- B.** A Timer can only execute following its defined schedule.
- C.** A Timer can only have the same schedule across all stages.
- D.** A Timer schedule can only be defined in the ODC Portal.

19. What is the purpose of the Transparency Service Pattern?

- A.** To provide a single service definition.
- B.** To abstract different systems into a single service.
- C.** To provide transparency in the routing of requests.
- D.** To provide different systems with the same API signatures.

20. What happens when the number of triggered Events exceeds the consumer app's capacity for parallel execution?

- A.** The Events will continue being added to its runtime queue until its capacity is reached.
- B.** The application pauses event processing and waits until capacity becomes available before resuming.
- C.** The processing time for currently executing Events is automatically reduced to accommodate the new Events.
- D.** The excess Events are discarded to prevent overloading the system.

21. Which of the following steps does the process of identifying bounded contexts involve?

- A. Disclosing business concepts
- B. Grouping concepts based on dependency clustering
- C. Determining sponsorship and ownership
- D. Understanding the teams, business owners, and sponsors

22. What is the recommended way to share business plain data between apps, knowing the data evolves frequently with business requirements?

- A. Encapsulating the business rules in Service Actions and referencing them.
- B. Referencing Static Entities directly.
- C. Encapsulating the business rules in Server Actions and referencing them.
- D. Referencing public Entities directly.

23. What happens behind the scenes when an ODC app is published?

- A. The app is compiled and deployed to the IIS server.
- B. The app is packaged into a container.
- C. The app is packaged into a virtual machine.
- D. The app is deployed to the next stage in the delivery pipeline.

24. Why is identifying Non-functional Requirements (NFRs) a fundamental step in the Architecture Design Process?

- A. Because NFRs guide the overall project timeline and priorities.
- B. Because NFRs define specific user tasks and features.
- C. Because NFRs describe the system's operational capabilities and constraints.
- D. Because NFRs describe software modules and features' behavior.

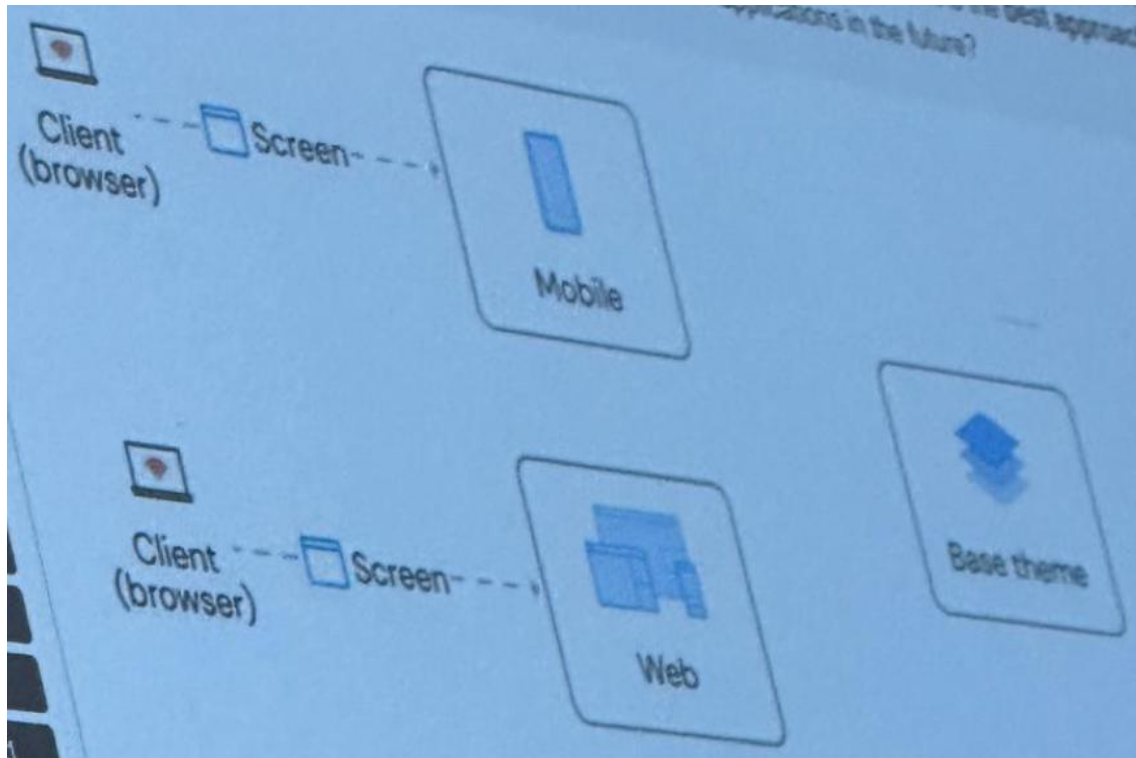
25. Consider a scenario where you need to fix a production issue in a REST integration being consumed in a Library. At which stage should the fix be applied?

- A.** In the Quality Assurance stage
- B.** In the Production stage
- C.** In the Development stage
- D.** In all stages simultaneously

26. What does it mean when we refer to 'lower cognitive load' in the context of architecture assembly?

- A.** It means we are promoting a design that will help increase the release cycle frequency.
- B.** It means we are decreasing the mental effort required to understand and work with the Architecture Design.
- C.** It means we are adding more complexity to the architecture design.
- D.** It means we are promoting a design that enables the creation of simpler code in the applications.

27. Consider a scenario where a mobile application and a web application will share some common look & feel defined in a Base Theme. Each application will then require specific look & feel customizations. What is the best approach to achieve this, knowing that the new Themes can be shared with more applications in the future?



- A.** Create two new Libraries, one for the Web Theme and another one for the Mobile Theme, to replace the Base theme. Then, each Theme will have its own Style Sheet with the required CSS defined. The Web and Mobile apps then reference the respective Themes.
- B.** Create two new Themes in the Library with the Base Theme. Then, each Theme references the Base Theme, inheriting the common CSS, and implements the specific and required CSS. The Web and Mobile apps then reference the respective Themes.
- C.** Create two new Libraries, one for the Web Theme and another one for the Mobile Theme. Then, each Theme references the Base Theme, inheriting the common CSS, and implements the specific and required CSS. The Web and Mobile apps then reference the respective Themes.
- D.** Create one Theme in the Web application and another one in the Mobile application. Then, each of these new Themes references the Base Theme directly, inheriting the common CSS, and implementing the specific and required CSS.

28. Considering the Architecture Design Process, which of the following is a benefit of defining bounded contexts?

- A. Improved collaboration between teams
- B. Clear and well-defined apps' ownership
- C. Simplified data model definition
- D. Reduced complexity and cognitive load

29. In which scenario should a Direct Integration be used instead of a Cold Cache?

- A. When frequent data updates are required
- B. When the source API does not support differential synchronization
- C. When the app cannot tolerate data update delays
- D. When app performance is not a concern

30. In some scenarios, referencing elements between apps and Libraries automatically makes a Structure public. Which of the following options does NOT describe one of those scenarios?

- A. Using Structures as parameters in public Server Actions defined in Libraries.
- B. Using Structures as parameters in public Client Actions defined in Libraries.
- C. Using Structures as parameters in public Service Actions defined in apps.
- D. Using Structures as parameters in public Server Actions defined in apps.